

Operator-1: Local Graph Expansion Operator — Full Explanatory Document

Author: Yue Lu
Version: 1.0

This framework is constructed based on Status-Relational Entropy (SRE) Dynamics.
All theoretical materials of this framework are archived in the Zenodo open-access repository. Except for the closed-source commercial engineering implementation modules corresponding to Operators 7, 8, 9 and 10, the mathematical-specification papers, algebraic derivations and simulation codes for Operators 1-6 are fully open-source. You may also access the fully open-source Tencent Smart Document with AI-assisted support (accessible via PC and WeChat mobile client). As of August 14, 2026, due to Google Terms-of-Service restrictions, the author no longer maintains or updates the SRE-series document library on Google Gemini notebook:

[Knowledge-Seeking] https://docs.qq.com/space/DUkRjYUtNWFdyV253?nlc=1&mode=ai_mode

According to the Status-Relational Entropy (SRE) principle, the foundations of classical physics originate from information statistics.

Document purpose: supporting notes for papers, writing reference, helping readers rapidly understand $\mathcal{G}_{n \rightarrow n+1}$.

Corresponding original reference:
Operator 1: Mathematical Specification and Algebraic Foundations of the Local Graph Expansion Operator (Yue Lu, v1.0)
<https://doi.org/10.5281/zenodo.21454140>, belonging to the Status-Relational Entropy Dynamics (SRE) theoretical system.

Core positioning: Operator-1 is neither a graph generator nor a dynamics solver. It serves as the underlying algebraic syntax layer of the SRE system, defining how the relational possibility-space expands with evolutionary pulses.

Table of Contents

- 1. Overview and Core Positioning
- 2. Foundational Axioms and Notation System
- 3. Mathematical Definition and Block-Matrix Structure of the Operator
- 4. Inductive-limit Ring R_∞ and Evaluation Homomorphism Φ (Separation between Symbolic Layer and Real-State Layer)
- 5. Key Mathematical Properties of the Operator (Theorem 2: Injectivity and Non-Surjectivity)
- 6. Connections of the Operator to Topological Frustration and the Diagonal Invariance Theorem
- 7. Operator-1 vs. Simulation Dynamics: Strict Demarcation between Operator Layer / Downstream Evaluation-Simulation Layer
- 8. Ontological Implications (SRE Relational Ontology)
- 9. List of Common Misconceptions
- 10. Physical Meaning within the SRE Cosmological Picture

1. Overview and Core Positioning

The local graph expansion operator $\mathcal{G}_{n \rightarrow n+1}$ implements dimensional expansion of the system from order n to order $n + 1$.

🚫 Critical distinctions:

1. The operator only constructs formal symbolic matrices; it performs no assignment, introduces no randomness, and computes no dynamical evolution.
2. It merely expands the potential relational possibility-set. Mapping symbolic variables to numerical real-world values -1 , $+1$, asynchronous activation, probabilistic evolution, and global feedback all belong to the downstream SRE dynamical module and are **not** part of Operator-1 itself.

Underlying ontological premise:

There is no pre-given geometric space or coordinates within the system. Graph vertices are merely index labels; vertices carry no intrinsic scalar values. All physical information is encoded in relations between vertices (bidirectional edges and vertex self-loops).

Each invocation of the operator:

- Adds one abstract vertex index;
- Generates a fresh set of algebraically independent formal symbolic variables representing coupling relations between the new vertex and all historical vertices, together with the self-loop relation of this new vertex;
- Strictly inherits the full historical structure, and never modifies old relations.

Core question addressed by Operator-1:

In a discrete system with only relations and no pre-existing space, what rigorous algebraic rules shall govern the expansion of the whole system's relational possibility-space, while guaranteeing lossless preservation of historical information and self-consistent temporal evolution.

2. Foundational Axioms and Notation System

Foundational Axioms (SRE System)

1. **Binary-Relational Range Axiom:** After evaluation via the homomorphism Φ , all relations (edges, self-loops) may only take values $+1$, -1 . Zero-values do not exist; there is no "no-edge" state. $+1$ denotes a cooperative relation; -1 denotes an antagonistic relation.
2. **Initial-Seed Axiom:** Evolution commences from the first-order matrix $M_1 = [1]$, representing the self-loop relation of the initial unit.
3. **History Read-Only Axiom:** Historical matrix sub-blocks, once generated, cannot be rewritten by the expansion operator.

Main Notation

Symbol	Meaning
$\mathbb{N}^+$	Set of positive integers 1, 2, 3, ..., excludes zero

Symbol	Meaning
J_n	Index reference set $1, 2, \dots, n$; 1-indexing is used throughout this document
$\mathcal{M}_n$	Space of n -dimensional discrete instance matrices: real-symmetric binary matrices with entries $\in \{-1, +1\}$
$\mathcal{M}_{n+1}[v_{n+1}]$	Parameter-matrix space carrying formal symbolic variables
v_{n+1}	Set of newly-injected independent formal variables at expansion step $n \rightarrow n + 1$: $x_{n+1,1}, \dots, x_{n+1,n}, y_{n+1}$
$\mathcal{R}_\infty$	Universal multivariate polynomial ring constructed via inductive limit, containing all formal variables over the full evolutionary history
Φ	Global evaluation homomorphism: maps formal variables inside $\mathcal{R}_\infty$ onto real numbers, subject to $\Phi(\cdot) \in \{-1, +1\}$
$\mathcal{G}_{n \rightarrow n+1}$	Local graph expansion operator
$\partial_{n \rightarrow m}$	Canonical embedding operator, embedding lower-order polynomial rings into higher-order polynomial rings

3. Mathematical Definition and Block-Matrix Structure of the Operator

Operator mapping:

$$\mathcal{G}_{n \rightarrow n+1} : \mathcal{M}_n \rightarrow \mathcal{M}_{n+1}[x_{n+1}, y_{n+1}]$$

Input: a fully-instantiated n -order real-symmetric binary matrix M_n (historically frozen real-world state).

Output: an $n + 1$ -order formal symbolic block matrix:

$$M_{n+1}(x_{n+1}, y_{n+1}) = \mathcal{G}_{n \rightarrow n+1}(M_n) = \begin{pmatrix} M_n & x_{n+1} \\ x_{n+1}^\top & y_{n+1} \end{pmatrix}$$

where

$$x_{n+1} = \begin{bmatrix} x_{(n+1,1)} \\ x_{(n+1,2)} \\ \vdots \\ x_{(n+1,n)} \end{bmatrix}$$

- $x_{(n+1,m)}$: formal symbolic variable for the coupling relation between the new vertex and old vertex m .
- y_{n+1} : formal symbolic variable for the self-loop relation of the new vertex.

Constraint: the top-left sub-block satisfies $M_{n+1}[1 : n, 1 : n] = M_n$; it is strictly inherited from the input and cannot be modified.

Storage aspect: upon expansion from n to $n + 1$, the array gains $2n + 1$ new matrix cells. Owing to matrix symmetry, only $n + 1$ of these carry independent algebraic degrees-of-freedom (n pairwise-coupling edges plus one self-loop). The remaining cells are symmetric mirror copies and carry no new information.

Minimal example: $M_1 \rightarrow M_2$

$$\mathcal{G}_{1 \rightarrow 2}(M_1) = \begin{pmatrix} 1 & x_{(2,1)} \\ x_{(2,1)} & y_2 \end{pmatrix}$$

One observes that the historical seed $M_1 = [1]$ is fully preserved; two independent symbolic variables are added and remain unassigned.

4. Inductive-limit Ring $\mathcal{R}_\infty$ and Evaluation Homomorphism Φ

Repeated invocations of $\mathcal{G}_{n \rightarrow n+1}$ continuously produce new formal variables. To avoid naming conflicts among variables, the universal polynomial ring $\mathcal{R}_\infty$ is constructed via the inductive (direct) limit within commutative algebra:

1. Variable sets generated in distinct expansion steps are pairwise disjoint: $v_k \cap v_m = \emptyset$ for $k \neq m$.
2. Canonical embeddings $\partial_{n \rightarrow m}$ realize injection of lower-order polynomials into higher-order rings.
3. $\mathcal{R}_\infty = \varinjlim \mathcal{R}_n$, containing all formal-symbol polynomials across the full evolutionary history.

$\mathcal{R}_\infty$ represents the complete relational possibility-space; at this stage all quantities remain purely symbolic with no assigned physical numerical values.

The global evaluation homomorphism Φ is the mapping transporting us from the symbolic layer toward realized physical states:

$$\Phi: \mathcal{R}_\infty \rightarrow \mathbb{R}$$

It satisfies homomorphism properties:

$$\Phi(f + g) = \Phi(f) + \Phi(g), \quad \Phi(f \cdot g) = \Phi(f) \cdot \Phi(g),$$

subject to the hard constraint:

$$\Phi(x_{(k,m)}) \in \{-1, +1\}, \quad \Phi(y_k) \in \{-1, +1\}.$$

Important notes:

- Operator-1 adds fresh formal variables into $\mathcal{R}_\infty$ (expanding the possibility-space).
- Φ projects possibilities onto realized numerical matrix states.
- Operator-1 itself does **not** perform evaluation via Φ .

5. Key Mathematical Properties of the Operator (Theorem 2: Injectivity and Non-Surjectivity)

1. Strict Injectivity

Given two distinct historical matrices $M_n^{(1)} \neq M_n^{(2)}$, the symbolic-matrix outputs of the operator are guaranteed to differ:

$$\mathcal{G}_{n \rightarrow n+1}(M_n^{(1)}) \neq \mathcal{G}_{n \rightarrow n+1}(M_n^{(2)}).$$

Physical interpretation: historical heterogeneity is preserved without loss; expansion never erases pre-existing structural information.

2. Strict Non-Surjectivity

The image set of the operator constitutes only a subset of the target symbolic-matrix space. The top-left $n \times n$ sub-block of any output must be a constant numerical block and cannot contain formal variables.

That is: not every $n + 1$ -order symbolic matrix can be generated by Operator-1.

Physical interpretation: the operator only expands new potential relations on top of already-fixed history. It cannot arbitrarily construct symbolic matrices lacking any frozen historical sub-structure.

6. Connections of the Operator to Topological Frustration and the Diagonal Invariance Theorem

Operator-1 supplies the formal matrix structure upon which the full set of core structural theorems can be derived.

Theorem 3: General Diagonal Invariance Theorem (valid only for matrix square M^2 , two-step graph walks)

Take the diagonal entry corresponding to the new vertex after formally squaring the expanded matrix:

$$(M_{n+1}^2)_{n+1,n+1} = \sum_{m=1}^n x_{(n+1,m)}^2 + y_{n+1}^2$$

After applying evaluation homomorphism Φ and using the identity $\forall s \in \{-1, +1\}, s^2 = 1$, we obtain:

$$\Phi((M_{n+1}^2)_{n+1,n+1}) = n + 1$$

This diagonal interference term evaluates to the constant $n + 1$ regardless of variable assignments.

⚠️ Caveat: this invariance holds **only for matrix squares**. It fails for higher-order matrix powers with $k \geq 3$.

Topological Frustration (Theorem 1, Corollary 1)

Topological frustration is defined **after** symbolic variables are evaluated to ± 1 and yield real-valued matrices:

1. Frustration of a single loop: the product of edge weights along a closed circuit equals -1 .
2. Global frustration criterion for complex networks: the system is globally frustration-free only if the product over all generators of the generalized cycle space (fundamental loops plus all self-loops) equals $+1$.

Operator-1 continuously creates new vertices and new relational slots, thereby generating new loops and new self-loops. Whether topological frustration emerges depends on the concrete assignments delivered by downstream Φ ; frustration is not an intrinsic property of the operator itself.

7. Operator-1 vs. Simulation Dynamics: Strict Demarcation between Operator Layer / Downstream Evaluation-Simulation Layer

This is the most frequently confused section; a sharp separation must be maintained.

Item	Operator-1 (Operator / Symbolic-Algebra Layer)	Hierarchical Dissipative Self-Organizing Binary-Network Dynamics (Downstream Dynamics & Simulation Layer) https://doi.org/10.5281/zenodo.20576606
Core Behaviour	Produces symbolic block matrices carrying formal variables	Executes evaluation homomorphism Φ , instantiates symbolic variables to ± 1
Stochasticity	Zero intrinsic randomness	Introduces stochastic dormancy via asynchronous activation χ_{ij}
Probabilistic Rules	None	$p_{ij}^{(n)}$ is jointly modulated by local frustration energy and evolutionary distance
Self-loop Handling	y_{n+1} remains an unassigned symbolic variable	Bottom-right self-loop is fixed by global negative feedback from total matrix sum; no randomness
Output Product	Polynomial symbolic matrix	Deterministic real-symmetric binary numerical matrix
Role	Algebraic syntax for evolution; expands possibility-space	Filters realized physical states out of the possibility-space

In simulation code, each iteration performs two actions simultaneously: it implements the structural expansion equivalent to Operator-1, and immediately executes downstream dynamics to assign values to all variables. For this reason many readers mistakenly treat simulation behaviour as intrinsic behaviour of the operator, which constitutes a typical misconception.

8. Ontological Implications (SRE Relational Ontology)

1. Vertices function purely as index markers; vertices possess no intrinsic values. All ± 1 values belong to relations (edges, self-loops). The self-loop y_{n+1} represents a “relation of a vertex toward itself”, and is **not** a numerical property of the vertex.
2. No pre-existing background geometric space is presupposed. Space, distance and dimensionality are not fundamental primitives; they are macroscopic emergent phenomena in the large-system statistical limit.
3. Each pulse iteration of Operator-1 advances fundamental evolution. Pulse-driven expansion itself constitutes the primitive manifestation of emergent time. New vertices are not dynamically generated from within the old network; they appear as the relational possibility-space expands as evolution proceeds.

Under current version: vertex generation follows the evolutionary-pulse axiom. Endogenous generation driven by internal system conditions remains a target for future work.

9. List of Common Misconceptions

Misconception 1: Operator-1 is an ordinary growing-graph algorithm (e.g. Barabási-Albert scale-free networks).

✗ False: conventional network algorithms directly output concrete graphs. Operator-1 only outputs symbolic possibility templates without value-assignment. Graph generation and stochastic evolution belong to downstream dynamics.

Misconception 2: New vertices are dynamically produced by the internal dynamics of the older network.

✗ False: internal dynamics only modify relational values. Vertex indices are introduced via the evolutionary-pulse axiom; they are not automatically triggered by internal tensions (within the current version).

Misconception 3: Diagonal matrix entries represent intrinsic state values of vertices.

✗ False: diagonal entries encode self-loop relations — a special class of relation — not intrinsic vertex properties.

Misconception 4: The operator itself includes randomness, dormancy rules and global feedback.

✗ False: all those mechanisms belong exclusively to the downstream SRE dynamical module.

Misconception 5: The $2n + 1$ newly-added matrix array cells correspond to $2n + 1$ new graph vertices.

✗ False: exactly one graph vertex is added per expansion step. $2n + 1$ counts storage-level array cells of the symmetric matrix; only $n + 1$ of those correspond to independent degrees-of-freedom.

Misconception 6: Topological frustration is an inherent property of Operator-1.

✗ False: topological frustration only arises **after** variable evaluation via homomorphism Φ . Frustration cannot be judged while variables remain purely symbolic.

10. Physical Meaning within the SRE Cosmological Picture

The full SRE framework — Status-Relational-Entropy Dynamics — maps onto Operator-1 as follows:

1. **Relational:** Operator-1 expands the relational possibility-space and constructs $\mathcal{R}_\infty$.
2. **Status:** Downstream evaluation homomorphism plus hierarchical dissipative dynamics select concrete realized system states out of the possibility-space.
3. **Entropy:** Entropy quantifies the statistical gap between the enormous latent set of relational possibilities and the small subset of actually-realized configurations. Asynchronous dormancy and topological frustration jointly drive entropy evolution.
4. **Dynamics:** Governs how realized states evolve atop the continuously expanding possibility-space.

Within the axiomatic system of SRE, the framework offers a conceptually self-consistent dissolution of the classic cosmological paradox: “What lies beyond the boundary of the universe?”.

The evolutionary boundary of the universe corresponds to the expansion frontier of Operator-1. Beyond this frontier there exists no space and no void. Only formal relational possibilities not yet realized by the evaluation homomorphism reside there. Extrapolating emergent geometric concepts of spacetime past this frontier constitutes an invalid logical extrapolation.

11. Open Theoretical Boundaries and Future Extensions

1. **Endogenization of vertex generation:** In the present version new vertices are introduced via an external-pulse axiom. Future work aims to trigger vertex birth internally by thresholds of

topological-frustration tension, path-interference effects or entropy fluctuations, eliminating the external-pulse postulate.

2. **Integration with Random-Matrix Theory (RMT):** Perform spectral-statistical analysis for this iteratively-grown matrix ensemble derived from the operator. Study universal large- N spectral behaviour and distinguish microscopic fluctuations from macroscopic statistical determinacy.
3. **Reduction of axioms:** Attempt to reduce the expansion axiom and binary-range axiom more deeply from the ontological premise that “a perfect zero (absolute nothingness) cannot be physically realized”.
4. **Generalization of the operator:** Extend operator formalism to permit non-uniform, non-sequential advance of the possibility-space frontier.

Supplementary Explanatory Note | This is interpretive material, not original mathematical specification. All mathematical proofs should be consulted in the original source paper.

DOI: <https://doi.org/10.5281/zenodo.21454140>
